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1. Cloud computing Fundamentals

Cloud computing

- Delivery of computing services over the internet

Characteristics on-demand access.

* Broad network access
* Resource Pooling
* Rapid elasticity.
* Scalability
* Measured service

Benefits:

* cost savings.
* flexibility.
* easy access.
* scalable resources.

Example:-

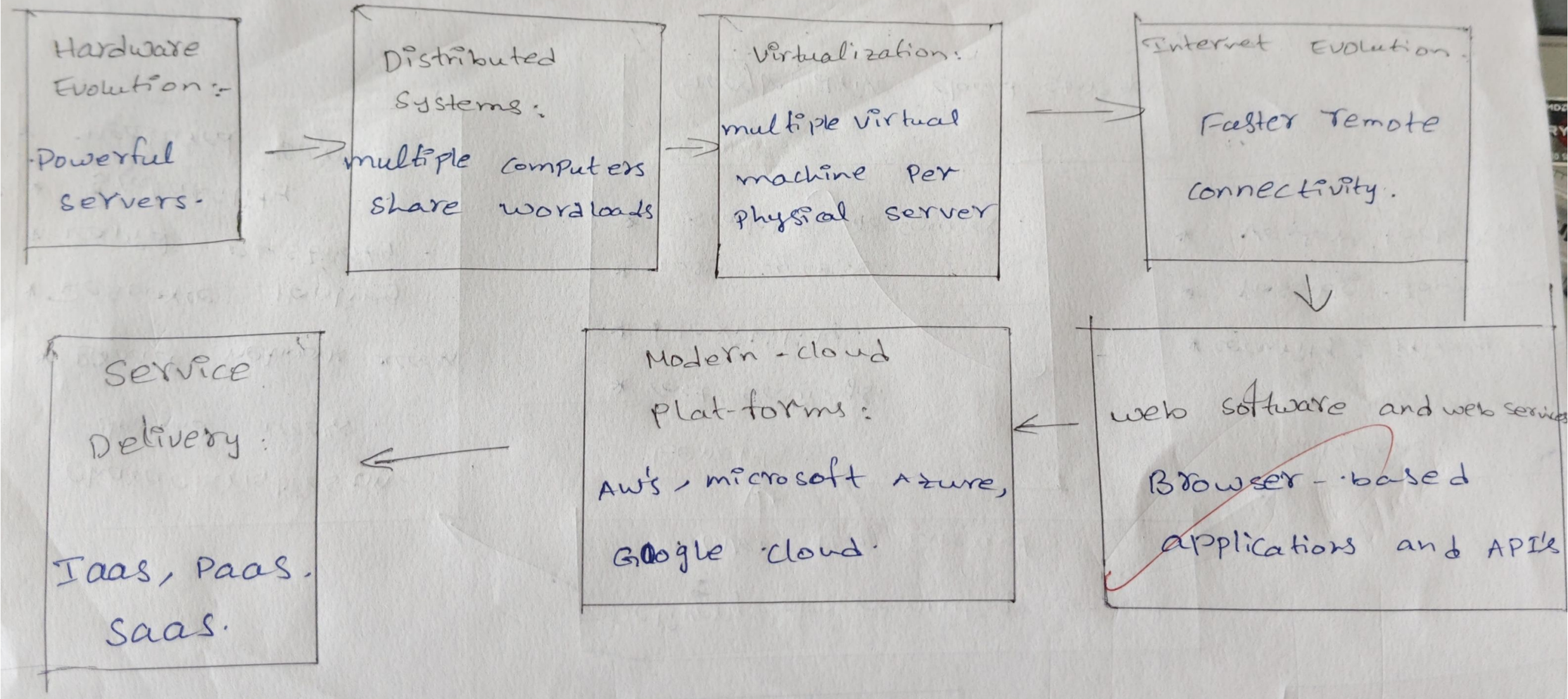
Google drive - cloud storage

limitations:

Internet dependency

* security & privacy risk.
* Possible downtime.
* Vender lock-in.

2. Evolution of cloud computing



3. cloud service Models.

Iaas:

Infrastructure layer's

Provider manager's.

Hardware, network

Virtualization

user manager:

OS, applications data

Paas:

platform layer provider

manages: * infrastructure,

OS, runtime / platform

user manages:

* Applications data.

Saas:

Application layer

* application & underlying infrastructure

* user software & manages content / settings.

Xaas:

Anything as a service

* services, (storage, security, software, etc....)

* service usage and configuration

Provider

Infrastructure & platform

→

Shared responsibility's

Depends on service model

←

Consumer.

applications and data.

4. Virtualization and webservices as Enablers of cloud.

Virtualization + web services

Virtualization

Physical server
creates / manage
↓
Hypervisior creates
↓
Virtual machines
Abstracts
↓
Abstraction and efficient
Resource use

web services

API
Enables communication
↓
SOAP and REST
↓
Application-to-service communication.

cloud deployment and Scalability

Delay Applications and resources

Increase / decrease resources as demand changes

provider Supplies service

Consumer access / uses service.